THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** acidic	Sour taste is a property of acids.
2	**(c)** an electric cell	A cell has chemicals that produce electric current.
3	**(d)** lustre	Metals reflect light.
4	**(a)** physical change	Only form or state changes.
5	**(a)** sulphuric acid	Sulphuric acid is a strong mineral acid.
6	**(d)** conductors	E.g. copper.
7	**(c)** brittle	They break when hammered.
8	**(d)** melting of wax	Wax solidifies again.
9	**(a)** base	Bases turn red litmus blue.
10	**(b)** hot	Heating effect of current.
11	**(a)** metal oxide	E.g. magnesium oxide.
12	**(a)** air and moisture	Both oxygen and water.
13	**(b)** basic	Bases are bitter and slippery.
14	**(a)** battery	A battery is a combination of cells.
15	**(a)** malleability	E.g. aluminium foil.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Substances that show the nature of a solution by colour change (1); Litmus and turmeric (1).	1+1
17	A complete path for the flow of current (1); It has a source, a switch and a device (1).	1+1
18	Lustre and malleability (1); Ductility and conductivity (1).	1+1
19	No new substance is formed (1); Melting of ice, tearing paper (1).	1+1
20	Baking soda (1); Lime water or soap (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Dip blue and red litmus (1); Red from blue means acid; blue from red means base (1); No change means neutral (1).	1+1+1
22	The circuit is open (1); No current flows (1); The filament does not heat (1).	1+1+1
23	Make a circuit with a gap (1); Place metal and non-metal samples (1); Metals light the bulb; non-metals do not (except graphite) (1).	1+1+1
24	Melting: physical (1); Burning and rusting: chemical (1); Boiling: physical (1).	1+1+1
	OR: Some are corrosive (1); Some are poisonous (1); Use indicators instead (1).	1+1+1
25	Wire melts on excess current (1); Circuit breaks (1); Appliance saved (1).	1+1+1
	OR: Malleable (1); Good heat conductor (1); Does not corrode easily (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Collect lemon juice, vinegar, soap, baking soda, salt (1); Test with litmus papers (1); Test with turmeric (1); Test with China rose (1); Record in a table and classify (1).	1+1+1+1+1
	OR: Make a circuit with a gap (1); Place objects in the gap (1); Copper wire, iron nail, coin: bulb glows (1); Rubber, plastic: no glow (1); Record in a table (1).	1+1+1+1+1
27	Lustre: gold (1); Malleability: aluminium (1); Ductility: copper (1); Conductivity: silver (1); Sonority: bell metal (1).	1+1+1+1+1

	OR: Physical: melting (1); Physical: tearing paper (1); Chemical: burning (1); Chemical: rusting (1); Chemical: souring of milk (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** chemical change	New substances appear.
2	**(b)** H_2SO_4	It contains two hydrogen atoms.
3	**(c)** insulators	E.g. rubber.
4	**(c)** oxygen	Oxygen is a gas.
5	**(a)** ripening of fruit	Cannot be undone.
6	**(b)** do not change the colour of litmus	They are neither acidic nor basic.
7	**(b)** electric iron	Heating element.
8	**(a)** basic	They turn red litmus blue in water.
9	**(b)** hydrated iron oxide	Reddish-brown flakes.
10	**(c)** lemon juice	Lemon juice contains citric acid.
11	**(b)** electric circuit	A complete path is needed.
12	**(a)** ductility	E.g. copper wire.
13	**(b)** physical change	Same substance, different state.
14	**(b)** $NaOH$	A common strong base.
15	**(a)** copper	Metals conduct.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Connect positive of one to negative of the next (1); Cells joined in series (1).	1+1
17	Mercury (1); Bromine (1).	1+1
18	Presence of oxygen (1); Presence of moisture (1).	1+1
19	HCl (1); $NaOH$ (1).	1+1
20	Cell: long and short parallel lines (1); Bulb: circle with a cross (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Mix them in a bottle (1); Gas formed (1); Pass through lime water (1).	1+1+1
22	Bacteria make acid from sugar (1); Acid damages enamel (1); Toothpaste is basic (1).	1+1+1
23	Voltages add (1); Bulb glows brighter (1); Battery formed (1).	1+1+1
24	Metal + acid gives salt + hydrogen (1); E.g. $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$ (1); Hydrogen burns with a pop (1).	1+1+1
	OR: Food gone bad smells (1); Milk turns sour (1); New substances (1).	1+1+1
25	Litmus: red in acid, blue in base (1); Turmeric: yellow in acid, red in base (1); China rose: dark pink in acid, green in base (1).	1+1+1
	OR: Rubber is an insulator (1); Circuit is not complete (1); No current (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Citric: lemon (1); Acetic: vinegar (1); Lactic: curd (1); Tartaric: tamarind (1); Ascorbic: amla (1).	1+1+1+1+1
	OR: Battery is two or more cells (1); Join + of one to - of next (1); 1.5 x 3 = 4.5 V (1); More voltage (1); Brighter bulb (1).	1+1+1+1+1
27	Mercury liquid (1); Sodium soft (1); Graphite conducts (1); Iodine lustrous (1); Diamond hard (1).	1+1+1+1+1
	OR: Colour change (1); Gas evolution (1); Heat and light (1); Precipitate (1); Smell change (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** bromine	Bromine is a reddish-brown liquid.
2	**(b)** burning of magnesium ribbon	New substance forms.
3	**(c)** base	Turmeric becomes red with soap solution.
4	**(c)** nichrome wire	High resistance wire that gets very hot.
5	**(d)** acidic	E.g. sulphur dioxide.
6	**(c)** painting	Coating blocks air and water.
7	**(b)** baking soda solution	Baking soda is a base.
8	**(a)** a closed circuit	The bulb glows.
9	**(d)** sonority	E.g. bells.
10	**(a)** chemical change	Ash and gases form.
11	**(b)** $\mathrm{Ca(OH)_2}$	It is a base.
12	**(a)** plastic	Plastic does not conduct.
13	**(b)** iodine	Iodine has a shiny appearance.
14	**(a)** stretching a rubber band	Shape only.
15	**(a)** acid	Acidic solutions turn it pink.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	It conducts electricity (1); It is soft and marks paper (1).	1+1
17	Fizzing (1); Carbon dioxide forms (1).	1+1
18	Acids turn blue litmus red (1); Bases turn red litmus blue (1).	1+1
19	Closed: current flows (1); Open: gap, no current (1).	1+1
20	Malleable: can be beaten into sheets (1); Ductile: can be drawn into wires (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Lustre: metals shiny, non-metals dull (1); Conductors vs insulators (1); Malleable vs brittle (1).	1+1+1
22	Three test tubes with nails (1); Dry air, boiled water with oil, moist air (1); Only moist air tube rusts (1).	1+1+1
23	Boil cabbage leaves and filter (1); Add to acid: turns red or pink (1); Add to base: turns green or blue (1).	1+1+1
24	High resistance (1); Gets very hot (1); Does not melt or oxidise easily (1).	1+1+1
	OR: Very reactive (1); Reacts with air and moisture (1); Kerosene protects it (1).	1+1+1
25	Natural: rain, seasons, day-night (1); Man-made: cooking, building, burning (1); Natural without human action (1).	1+1+1
	OR: Damage monuments (1); Harm plants and animals (1); Can be neutralised with bases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Acid + base reaction (1); $\mathrm{HCl + NaOH \longrightarrow NaCl + H_2O}$ (1); Salt and water formed (1); Treating acidity (1); Treating acidic soil (1).	1+1+1+1+1
	OR: Cell: gives energy (1); Wires: path (1); Switch: control (1); Bulb: glows (1); Closed circuit needed (1).	1+1+1+1+1
27	Non-metals brittle (1); Dull (1); Poor conductors (1); Not sonorous (1); Metals are opposite (1).	1+1+1+1+1
	OR: Iron + oxygen + water gives rust (1); Needs air and moisture (1); Paint (1); Galvanise (1); Alloy as stainless steel (1).	1+1+1+1+1

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